MIL-L-24282(SHIPS)
3 July 1968
SUPERSEDING
MS 19106 (SHIPS)
16 November 1961

MILITARY SPECIFICATION

LIGHT, SIGNAL, MULTIPURPOSE, KIT, NONMAGNETIC, SYMBOL 106.1

1. SCOPE

1.1 This specification covers a nonmagnetic, portable, multipurpose light, Symbol 106.1, complete with red, green, and yellow lenses and a stowage box for signaling, visual communications, and limited lighting for shipboard use. It can be operated by 115-volt, alternating current (a.c.) or by self-contained dry cell batteries.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-490	- Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
TT-P-664	- Primer Coating, Synthetic, Rust Inhibiting, Lacquer-Resisting.

MILITARY

MIL-R-11	- Resistors, Fixed, Composition (Insulated) General Specification.
MIL-M-14	- Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-B-18/1	- Batteries, Dry, BA-2.
MIL-R-900	- Rubber Gasket Material, 45 Durometer Hardness For Moderately Low Temperature Service.
MIL-S-901	- Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-E-917	- Electric Power Equipment, Basic Requirements for (Naval Shipboard).
MIL-D-1000/2	- Drawings, Engineering and Associated Lists.
MIL-R-2726/50	- Receptacle, Plug, Electrical, 15 Ampere, 125/250 Volts, 2 Pole Grounded (Symbol No. 720.3).
MIL-T-7928	- Terminals, Lug Splices, Conductor: Crimp Style, Copper.
MIL-P-8585	- Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity.
MIL-M-10578	- Metal Conditioner and Rust Remover (Phosphorus Acid Base).
MIL-P-15024	- Plates, Identification—Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-P-15037	- Plastic Sheet, Laminated, Thermosetting, Glass Cloth, Melamine Resin.
MIL-E-15090	- Enamel, Equipment, Light Gray (Formula No. 111).
MIL-P-15328	- Primer, Pretreatment (Formula No. 117 For Metals).
MIL-I-17214	- Indicator, Permeability; Low-Mu (Go-No-Go).
MIL-E-17555	- Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-S-19622	- Stuffing Tubes (Nylon), General Specification.
MIL-L-19896	- Labels and Label Tape, Pressure Sensitive Adhesive, Paper, Water Resistant.
MIL-I-45208	- Inspection System Requirements.

STANDARDS

MILITARY

MIL-STD-167	- Mechanical Vibrations of Shipboard Equipment.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MS3102	- Connector, Receptacle, Electric, Box Mounting.

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MS3106	- Connector, Plug, Electric, Straight.
MS 16569	- Switch, Toggle, D. P. S. T.
MS 16655	- Lampholder for Two-Prong Sealed Beam Lamp.
MS 16845	- Lamp, Incandescent, PAR 46, Mogul End Prong, Stippled Cover, Multi-purpose Signal.
MS 25100	- Switches, Toggle, Double Pole, One-Hole Mounting (Sealed Toggle Bushing).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL FIRE PROTECTION ASSOCIATION(NFPA)
70 - National Electrical Code.

(Application for copies should be addressed to National Fire Protection Association, 60 Batterymarch Street, Boston, Massachusetts 02110.)

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards for Federal Services.
Handbook 95 - Standard Colors of Signal Lights.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Blvd, Chicago, Illinois 60606.)

AMERICAN WELDING SOCIETY
Welding Handbook

(Application for copies should be addressed to the American Welding Society, Inc., United Engineering Center, 345 East 47th Street, New York, N. Y. 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to production, a sample manufactured by production tools and processes shall be tested as specified in 4.2 (see 6.3).

3.2 Design. The signal light kit shall be designed to conform to figures 1 through 4 and as specified herein. The light shall be suitable for signaling, visual communications, and limited lighting.

3.3 Construction. The signal light shall be constructed for long service without frequent overhaul under severe conditions encountered aboard ship. The construction shall be simple and sturdy and such that continued satisfactory operation in accordance with this specification and routine maintenance in service do not require special tools or excessively precise workmanship. Special tools are defined as those tools not listed in the Federal Supply Catalog. (Copies of this Catalog may be consulted in the Officer of the Defense Contract Administrative Service (DCAS)).

3.4 Materials. Materials shall be as specified herein, and shall meet the performance requirements specified herein. All materials shall be noncorrosive and nonmagnetic.

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3.4.1 Nonmagnetic. The magnetic permeability of all fixture parts shall be 2.0, or less, after fabrication. The transformer is excluded.

3.4.2 Plastics. Plastic materials shall be free of defects such as embedded particles, bubbles, scratches, blemishes, and striations.

3.4.3 Stress relief.

3.4.3.1 Metals. All metals used in fabrication and assembly shall be treated to prevent distortion or failure due to stresses or other conditions resulting from working, forming, and welding. Heating or peening after welding will accomplish stress relief (see American Welding Society Welding Handbook).

3.4.3.2 Plastics. Adequate measures shall be taken in molding and processing plastics to insure that stress buildup does not occur, or is satisfactorily treated to relieve these stresses to prevent deterioration or failure of the part or assembly.

3.4.4 Insulating materials. Insulating material shall be type GME of MIL-P-15037. Molding compounds shall conform to MIL-M-14.

3.4.5 Rubber. Material for rubber gaskets shall conform to MIL-R-900.

3.4.6 Dissimilar metals. Application and use of dissimilar metals shall conform to the requirements of MIL-E-917.

3.4.7 Restricted material.

3.4.7.1 Cast iron (gray iron) or other brittle material shall not be used.

3.4.7.2 Material listed in MIL-E-917 as prohibited shall not be used.

3.5 Threaded parts. Screw threads shall be UNC class 2 of Handbook H 28.

3.5.1 Inserts, tapping, and threaded devices. Inserts, tapping, and threaded devices shall be as follows:

(a) Self-tapping screws and sheet metal screws shall not be used.
(b) Molded-in-place inserts in plastic for screw threads are required.

3.5.2 Locking devices. Locking devices shall be used to secure mechanical assemblies to provide continuous operation under the conditions of shock and vibration specified herein.

3.6 Welding. Welding and allied processes used in fabrication shall be in accordance with the American Welding Society Welding Handbook.

3.7 Beam spread. The signal light shall be so designed to permit the following beam spread limits:

Horizontal - 7 degrees
Vertical - 7 degrees

3.8 Castings. Castings shall be free from cold shuts, blow holes, or any imperfections that may affect strength. They shall be finished smooth, that is, no openings or rough edges.

3.9 Dimensional tolerances. The following tolerances shall apply:

(a) Fractional ± 1/64 inch.
(b) Decimal ± 0.005 inch.
(c) Angular ± 1 degree.

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3.10 Electrical requirements.

3.10.1 Clearances. Minimum electrical creepage and clearances shall be as required by the National Fire Association Handbook 70, National Electrical Code.

3.10.2 Cable. Electrical cables shall be three-conductor. The green conductor shall be used for grounding purposes. Type and size shall be as specified herein.

3.10.3 Terminals. Terminals shall be in accordance with MIL-T-7928. Terminals shall be used on all inside light connections.

3.10.4 Grounding. Means shall be provided to attach the green conductor of the cable to the inner face of the light housing and stowage box. Grounding shall comply with figure 3. Precautions shall be taken to insure the safety of operational personnel to the maximum practicable extent.

3.10.5 Cable entrance. Cable entrance into the housing shall be by plastic stuffing tubes in accordance with MIL-S-19622.

3.10.6 Safety. Precautions shall be taken to insure the safety of operating personnel.

3.11 Performance.

3.11.1 Operation. The signaling light, when tested as specified in 4.4.1, shall operate properly in every respect. The voltage to the lamp, when operated from the stowage box power supply, shall be 20 volts ± 1 volt.

3.11.2 Dielectric withstanding voltage. There shall be no evidence of breakdown when the light assembly is subjected to the test specified in 4.4.2.

3.11.3 Insulation resistance. The insulation resistance between all current-carrying and noncurrent-carrying parts shall be not less than 1 megohm (see 4.4.3).

3.11.4 Shock. Stowage box, signal light, and components shall withstand, without damage or loosening of parts, the high impact (H.I.) shock test specified in 4.4.4.

3.11.5 Vibration. Stowage box, signal light, and components shall withstand, without damage or loosening of parts, the vibration test specified in 4.4.5.

3.12 Components.

3.12.1 Kit. Each kit shall contain one stowage box (see 3.12.9) which in turn shall contain the following:

(a) Complete light assembly (see 3.12.2).
(b) Lamp and gasket (see 3.12.3).
(c) Lenses one each red, green, and yellow (see 3.12.4).
(d) Extension cable (see 3.12.5).
(e) Power cable (see 3.12.6).
(f) Instruction manual (see 3.12.7).

3.12.2 Light assembly. The multipurpose, portable signal light shall consist of a device which produces a high - intensity beam of light suitable for use as a spotlight or as a blinker for visual communications when the beam is electrically interrupted by means of a trigger switch. The light shall provide for sighting and holding the beam on the desired target. The light shall be so designed to permit operation from self-contained batteries (see 3.12.8) or from a 115/20 volt, A.C. source of the stowage box (see 3.12.9). The general dimensions, configurations, and arrangement of the equipment shall be as shown on figures 1 and 2 and as specified hereinafter.

3.12.2.1 Housing. The housing shall consist of an aluminum cylinder closed at one end. The housing shall be so designed as to accommodate three batteries (see 3.12.8), switches (see 3.12.2.4), rear handle (see 3.12.2.5.2), cover (see 3.12.2.2), rear sight (see 3.12.2.3), and the internal wiring and hardware specified in 3.12.2.6.

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3.12.2.2 Cover. The aluminum cover shall be designed to accommodate the lens (see 3.12.4), lamp (see 3.12.3), front handle (see 3.12.2.5.1), and front sight (see 3.12.2.3). The cover shall be firmly attached to the housing by means of two latches.

3.12.2.3 Sights. Sights shall be provided as shown on figures 1 and 2, and shall be aligned to be parallel to the optical axis of the sealed beam lamp. The rear sight shall be a "V" notched type and the front beaded type.

3.12.2.4 Switches. Switches, generally, shall be mounted as shown on figures 1 and 2. A toggle switch double pole, double throw power transfer (On-Off-On) type in accordance with MS 25100, part number MS 25100-23, shall be provided to permit switching to an alternate power supply. A second-stage toggle, single pole, single throw, three by-pass switch, MS part number MS 16569-1, shall be connected across a 2-watt, 6.2-ohm resistor in accordance with MIL-R-11, connected as shown on figure 3, for dimming purposes.

3.12.2.5 Handles.

3.12.2.5.1 Front. The front handle shall be of aluminum, and shall be finished smooth, with no openings or rough edges. The handle shall be an adjustable type to assure a steady position while signaling. A device shall be provided for locking in the forward position. If necessary, the handle may be mounted off center.

3.12.2.5.2 Rear. The rear handle (pistol grip) shall be of either aluminum on molded plastic of a grade adapted for the purpose. The rear-handle shall contain a trigger operated switch capable of momentary contact such as would be required for signaling a light beam. The trigger switch shall have positive open and closed positions and, when operated at the rate specified in 4.4.6, it shall successfully complete 50,000 cycles of operation. A male receptacle, Part-Number MS 3102 R-10 SL-3 P in accordance with MS 3102, shall be securely mounted as shown on figure 1, and shall be connected as shown on figure 3.

3.12.2.6 Internal wiring. Internal wiring, Underwriters Laboratory approved No. 18 shall be provided and shall be connected as shown on figure 3. Connection to the lamp shall be made by a socket in accordance with MS-16655.

3.12.3 Lamp. The lamp shall be sealed beam, par. 46, Industry Number 4300, 20-volt in accordance with MS 16845. The lamp shall be fitted to a V-shaped rubber gasket, and shall be firmly mounted to the cover by means of an easily removable retaining ring.

3.12.4 Lens. Lenses shall be furnished in three colors, signal red, signal yellow, and signal green in accordance with Handbook 95. The lens shall consist of a plastic disc 5-3/8 inches in diameter and 3/32 inch thick. The lens shall be easily attached to the cover by means of three berylluim copper clips.

3.12.5 Extension cable. A cable assembly for supplying power to the signal light from the stowage box shall be furnished. This cable shall be 25 feet long, type UL-SWO (3/18) fitted with a plug, Part number MS 3106 R-10SL-3S in accordance with MS 3106 at one end and plug, Part Number MS 3106R-10SL-3P, at the other.

3.12.6 Power cable. A cable assembly for supplying power to the stowage box from the ship's service shall be furnished. It shall be 15 feet long, type UL-SVO (3/18), with Industry number 5269 female plug at one end and male plug, symbol number 720-3, in accordance with MIL-R-2726/50 on the other.

3.12.7 Instruction manual. An instruction manual shall be furnished for each complete signal light. The manual shall contain the following information: complete instruction for the care, operation, and maintenance of the signal light; parts identification by style, model, or catalog number as available; and complete wiring diagram.

3.12.8 Batteries. Batteries shall be type BA-2, 22 1/2 volt in accordance with MIL-B-18/1. Three batteries shall be connected in parallel. Unless otherwise specified (see 6.2), batteries need not be furnished by the contractor.

3.12.9 Stowage box. Stowage box in general shall be constructed as shown on figure 4. Overall size shall be kept to a minimum. The dimensions shall not exceed those shown on figure 4. Material shall be a

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good commercial grade of aluminum, protected against corrosion and of sufficient gage to provide firm support for component and compartmentation without excessive weight. A commercially available box and cover, adapted for the purpose, of drawn or fabricated construction will be acceptable.

3.12.9.1 Compartmentation. Compartmentation shall be provided for items specified in 3.12.1 aluminum baffles and cushions of foamed plastic, rubber, or felt shall be provided to prevent movement and damage of components in transit and stowage aboard ship. Kits shall conform to the requirements of 3.11.4 and 3.11.5.

3.12.9.2 Transformer. The transformer shall be 115/20 volt. The transformer shall be firmly mounted to the box and wired to receptacles as shown on the wiring diagram, figure 3.

3.12.9.3 Lamp receptacle. The receptacle for energizing the lamp shall be MS 3102 R-10SL-3P attached to the box adjacent to the transformer. A plastic cap attached to a safety wire anchored to the box shall be provided over the receptacle for protection.

3.12.9.4 Power supply receptacle. The power supply receptacle shall be industry number 5278, male, and shall be flush-mounted to the box adjacent to the transformer.

3.13 Weight. Component parts shall be as small and light as possible but sufficiently rugged to withstand rough handling so that mechanical damage or distortion will not be of sufficient magnitude as to impair continued operation. The total weight of the signal light shall not exceed 4-1/4 pounds without batteries. The total weight of the signal light kit, without batteries, shall not exceed 16-1/2 pounds.

3.14 Identification and switch plates. Identification and switch plates shall be in accordance with MIL-P-15024. Identification plates shall be as shown on figure 4, and shall be securely attached on the stowage box as shown on figure 4. Switch plates shall be as shown on figure 2 and shall be securely attached on the signal light as shown on figures 1 and 2.

3.15 Label. One label, type IV conforming to MIL-L-19896 shall be applied as shown on figure 1. It shall indicate the following information:

"USE LAMP FSN 9G, 6240-828-3039".

3.16 Drawings. Drawings shall be type II form 2 category F of MIL-D-1000/2. In addition, reference to MIL-L-24282 shall appear in the title block or as near as possible to the title block. Drawings used for manufacturing shall be processed as specified in MIL-D-1000/2.

3.16.1 Drawing distribution. The manufacturer, upon being notified that his first article inspection sample has passed the specified inspection, shall place on record a set of prints of the drawings used for manufacturing the signal light kit with the following activities:

(a) One set for file with the DCAS.
(b) One set for file with the Naval Ship Engineering Center Department of the Navy, Washington, D. C. 20360.
(c) One set for file with the contracting officer.

3.17 Protection against corrosion. In order to prevent corrosion, all metal parts shall be of corrosion-resisting materials or other materials treated in a manner to render them adequately resistant to corrosion. Treatment shall be in accordance with MIL-E-917.

3.18 Painting. All aluminum parts shall be completely painted in accordance with the requirements specified hereinafter. Plastic enclosures normally will not be painted.

3.18.1 Cleaning. After all machining, welding, and brazing operations are completed, the exterior and interior surfaces of all metal enclosures shall have all rust or other visible corrosion products removed; (rust removal from ferrous metal surfaces shall be in accordance with MIL-M-10578, and shall be thoroughly cleaned of all grease, oil, and dirt by solvent wiping, vapor degreasing, or caustic washing and rinsing.

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3.18.2 Undercoats. Either of the following undercoats shall be applied:

(a) One coat of primer pretreatment coating in accordance with MIL-P-15328 shall be applied as a continuous film 0.0002 to 0.0005 inch thick.
(b) A hot diptank phosphate or chromate-phosphate treatment conforming to type I of TT-C-490 followed by one coat of primer, conforming to TT-P-664 or MIL-P-8585 and applied as a continuous film 0.0002 to 0.001 inch thick.

3.18.3 Finish coats. Light gray enamel conforming to MIL-E-15090 shall be applied in sufficient thickness over the undercoat to provide protection of the equipment and to provide a satisfactory appearance.

3.19 Workmanship. The signaling light shall be manufactured and finished in a first class manner. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, machine screw assemblage, welding, freedom of parts from burrs and rough edges, and finish.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The supplier shall provide and maintain an inspection system in accordance with MIL-I-45208.

4.2 First article inspection. The first article inspection sample (s) shall be subjected to the examination specified in 4.3.3 and the tests of 4.4.

4.3 Quality conformance inspection.

4.3.1 Lot. For the purpose of sampling, a lot shall be the identical number of signal light kits on the contract or order.

4.3.2 Sampling for examination and tests. A random sample of signaling light kits shall be selected in accordance with table I from each lot for the examination of 4.3.3 and the tests specified in 4.3.4. Any signal light kit containing one or more defects or failing one or more tests shall not be offered for delivery. If the number of such defective light in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall not be offered for acceptance.

Table I - Sampling for examination and tests.

Number of Signal light kits in lot	Number of Signal light kits in sample	Acceptance number	Rejection number
2 to 10	All	-	-
11 to 15	10	0	1
16 to 25	13	0	1
26 to 40	17	0	1
41 to 65	22	0	1
66 to 110	28	1	2
111 to 180	35	1	2
181 to 300	45	2	3
301 to 500	55	2	3
501 and over	70	3	4

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4.3.3 Examination. The samples selected in accordance with 4.3.2 shall be examined to determine compliance with the requirements of this specification for material, design, construction, dimensions, weight, marking, and workmanship.

4.3.4 Tests. The sample(s) selected in accordance with 4.3.2 shall be subjected to the operation test, the dielectric withstanding voltage test, and the insulation resistance tests of 4.4.1, 4.4.2, and 4.4.3.

4.4 Test procedures.

4.4.1 Operation. The sample(s) shall be checked for operation with both self-contained d.c. power supply and the a.c. power supply from the stowage box.

4.4.2 Dielectric withstanding voltage. Dielectric withstanding voltage between electrical circuits and between electrical circuits and ground (equipment frame), shall be tested with a closely sinusoidal source of 60 cycles, a.c. in accordance with method 301 of MIL-STD-202. Root mean square (r.m.s.) value of the test voltage shall be 1000 volts. The lamp shall be disconnected from the signal light during this test.

4.4.3 Insulation resistance. The insulation resistance between electrical circuits and between electrical circuits and ground (equipment frame) following the dielectric tests specified in 4.4.2 shall be not less than the value specified in 3.11.3. Insulation resistance shall be measured with a megohmmeter at a voltage not less than 500 volts, d.c., at normal room temperature. The lamp shall be disconnected from the signal light during this test.

4.4.4 Shock. The stowage box, complete with component parts (see 3.12.1), shall be securely strapped to the test plate and subjected to the grade A, class II, type A shock test in accordance with MIL-S-901.

4.4.5 Vibration. The stowage box, complete with component parts (see 3.12.1), shall be subjected to the type I vibration test of MIL-STD-167.

4.4.6 Life. The trigger switch and mechanism shall be subjected to a life test of 50,000 cycles of operation at the rate of 15 cycles per minute to determine conformance with 3.12.2.5.2. One cycle is defined as one complete movement from the OFF to ON and back to OFF position.

4.4.7 Magnetic permeability. The magnetic permeability of material shall be determined with an indicator in accordance with MIL-I-17214, or equal.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Domestic shipment and early equipment installation.

5.1.1 Preservation and packaging. Preservation and packaging which may be the supplier's commercial practice shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation. (see 5.3).

5.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier, at lowest rate, and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation, and may conform to the supplier's commercial practice. (see 5.3).

5.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal, stock number or manufacturer's part number, contract or order number contractor's name and destination (see 5.3).

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5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation and packaging, packing, and marking for shipment shall be as specified by the procuring activity (see 6.2)

5.2.1 Domestic shipment and storage and overseas shipment. The signal light, repair parts, and technical publications shall be preserved and packaged level A, packed level A, B, or C and marked in accordance with MIL-E-17555 (see 5.3)

5.3 Use of polystyrene (loose-fill) material.

5.3.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.2), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labeled as follows:

"CAUTION

Contents cushioned, etc. with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiberboard or transparent flexible cellular material."

5.3.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler, and dunnage.

6. NOTES

6.1 Intended use. This kit is designed for use in the following services:

(a) Ship to ship signaling.
(b) Ship to plane signaling.
(c) Shore to ship signaling.
(d) Portable spotlight.
(e) Covert operation (kit can be modified for infrared capability).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Specify when battery is required to be furnished (see 3.12.8).
(c) Preservation and packaging, packing, and marking if other than specified in 5.1 (see 5.2)
(d) When polystyrene (loose-fill) material is approved.

6.3 First article inspection. Invitations for bid should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy-Sh
(Project 5850-NO18Sh)

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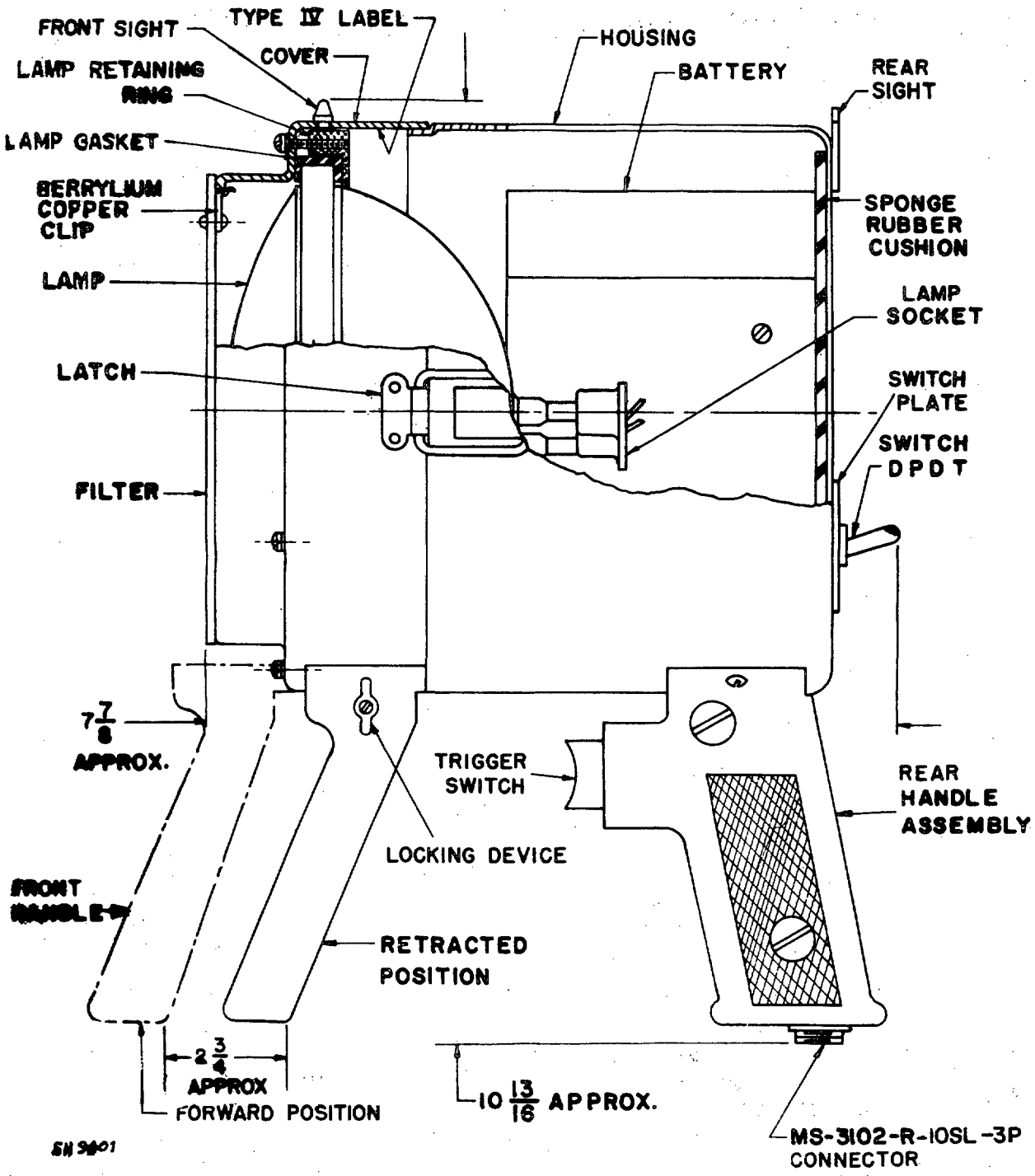


FIGURE 1. – LIGHT ASSEMBLY

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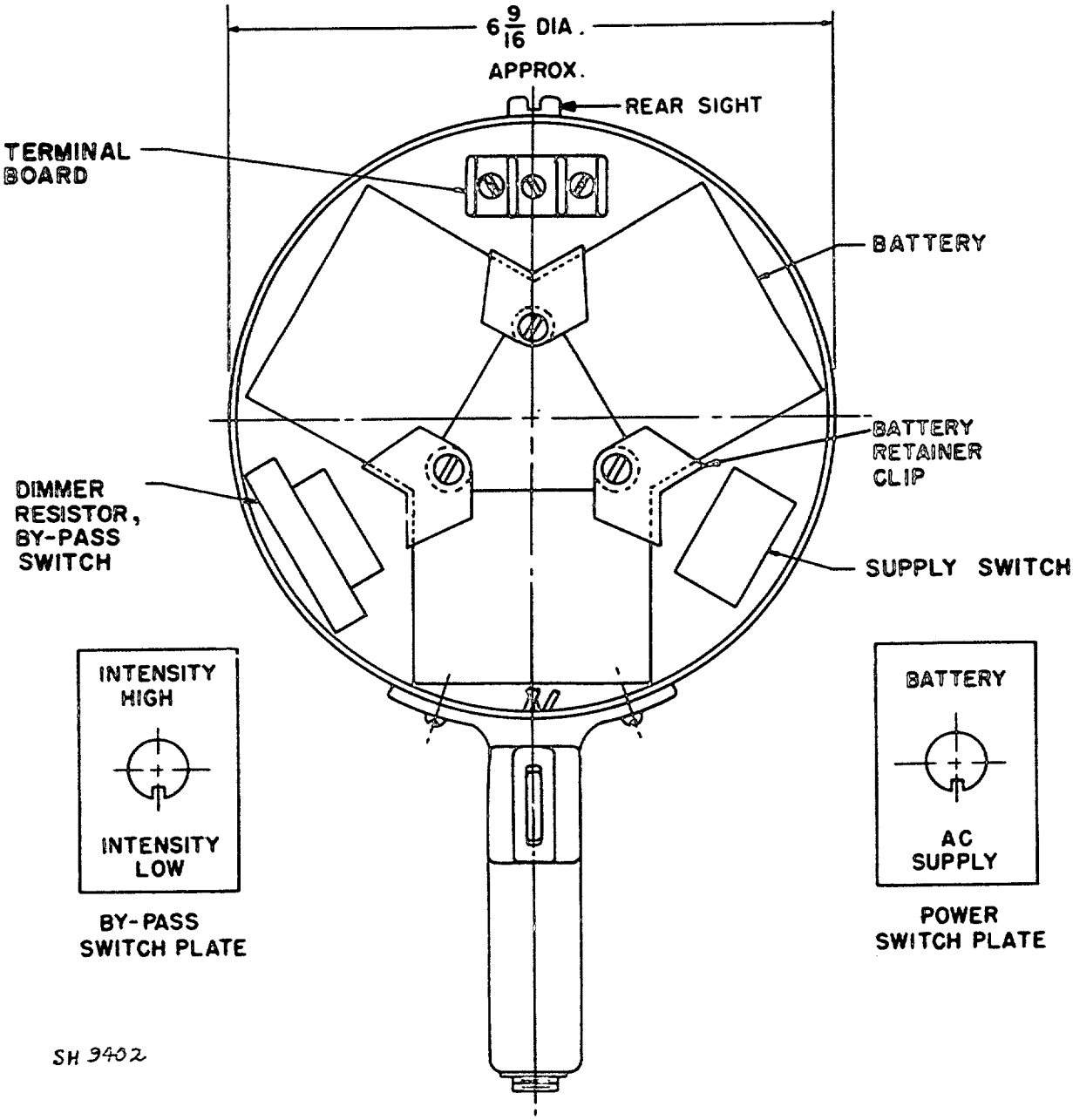


FIGURE 2. – LIGHT ASSEMBLY WITH COVER REMOVED

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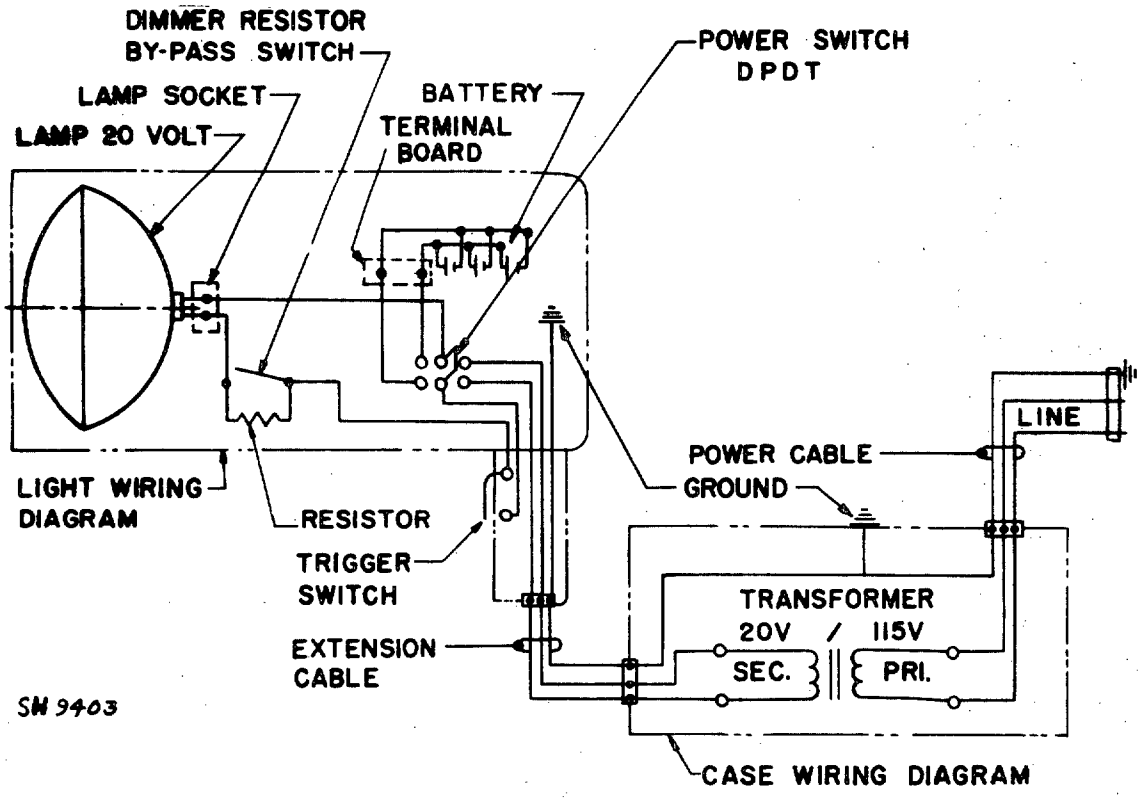
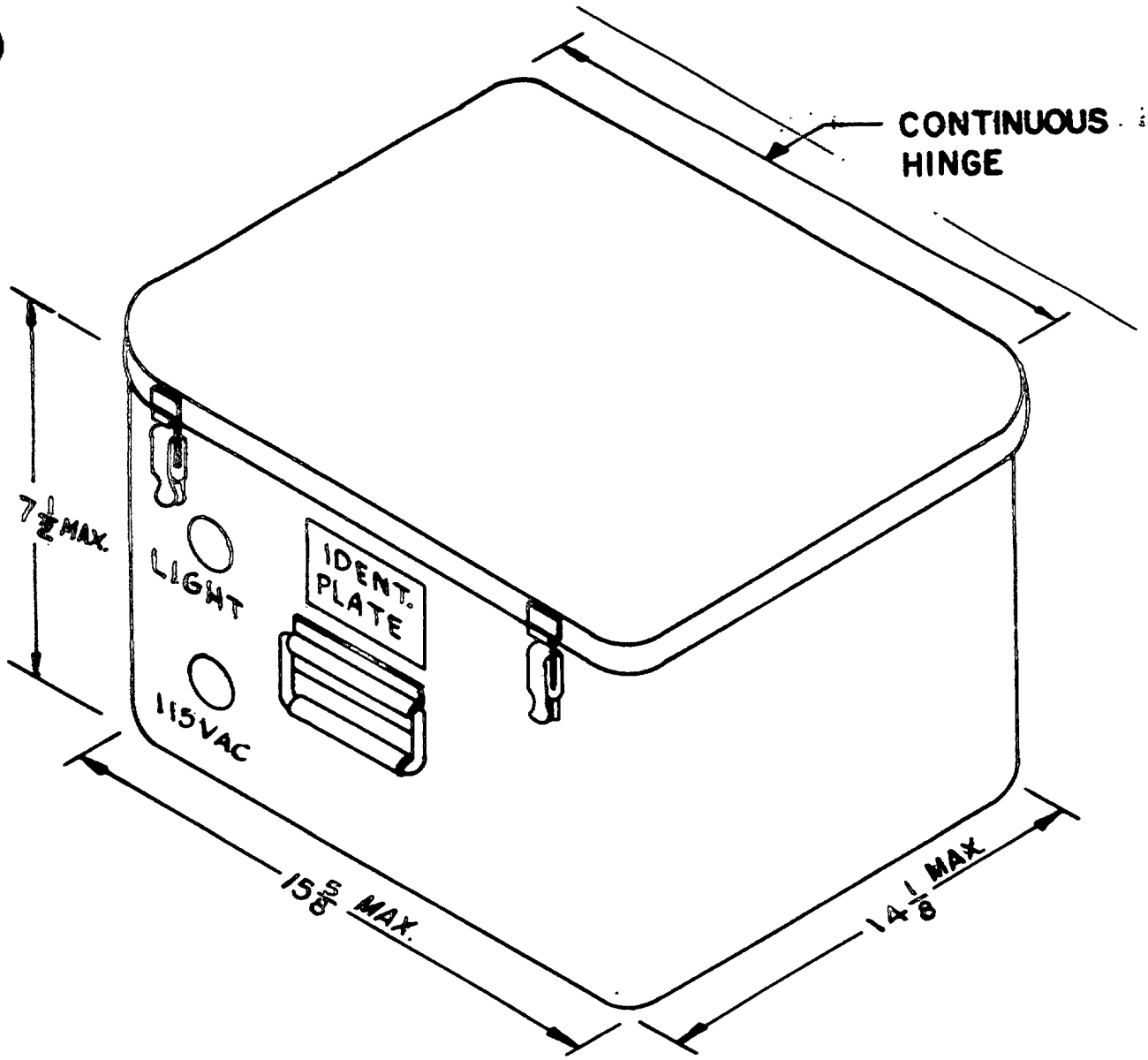


FIGURE 3. – WIRING DIAGRAM

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MULTIPURPOSE SIGNAL LIGHT KIT
SYMBOL 106.1, MIL-L-24282
CONTRACT NO._____ DATE (YEAR)
MANUFACTURER OR TRADEMARK

IDENTIFICATION PLATE

SH 9404

FIGURE 4. – STOWAGE BOX

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated.. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1 APR 63 1426 (NAVSHIPS 0VPRT 12-66)

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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